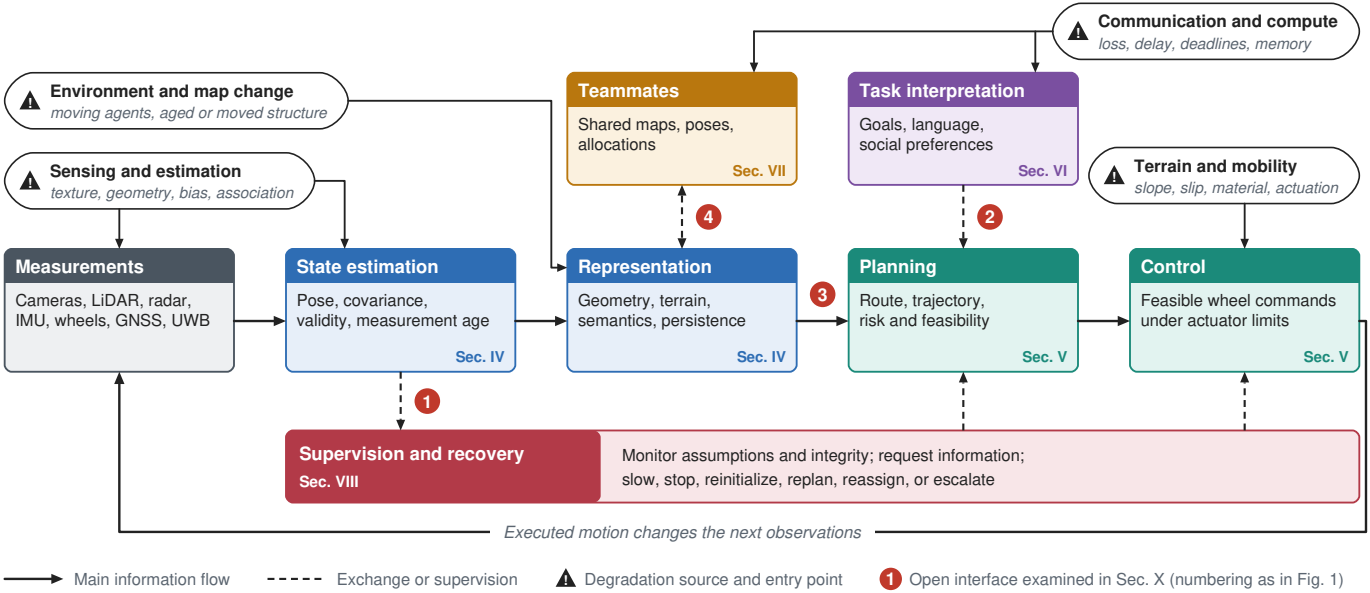

Communication and compute
loss, delay, deadlines, memory
Environment and map change
moving agents, aged or moved structure
Sensing and estimation
texture, geometry, bias, association
Teammates
Shared maps, poses, allocations
Sec. VII
Task interpretation
Goals, language, social preferences
Sec. VI
Terrain and mobility
slope, slip, material, actuation
Measurements
Cameras, LiDAR, radar, IMU, wheels, GNSS, UWB
State estimation
Pose, covariance, validity, measurement age
Sec. IV
Representation
Geometry, terrain, semantics, persistence
Sec. IV
Planning
Route, trajectory, risk and feasibility
Sec. V
Control
Feasible wheel commands under actuator limits
Sec. V
1
2
3
4
Supervision and recovery
Sec. VIII
Monitor assumptions and integrity; request information; slow, stop, reinitialize, replan, reassign, or escalate
Executed motion changes the next observations
Main information flow
Exchange or supervision
Degradation source and entry point
Open interface examined in Sec. X (numbering as in Fig. 1)